Project Report

Agent-Based Modelling and Financial
Price Prediction

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1 Introduction

1.1 General context

Financial markets are complex systems in which a large number of agents with heterogeneous strategies interact. Understanding and modelling these dynamics is a major challenge, both for risk management and for the design of efficient investment strategies. Agent-Based Models (ABM) offer a promising approach for capturing the emergent collective behaviours that elude classical equilibrium models.

This project is part of the course *ST5 — Time series and agent-based models in Finance*. It aims to explore the modelling of investor behaviour on a fictitious market (ZEL), and then to test the predictive power of the calibrated model on real data (NVIDIA, 2015–2025).

1.2 Objectives

The objective is to build a complete quantitative modelling pipeline, from model estimation to the evaluation of a trading strategy. We set ourselves the following goals:

- Estimate the parameters of a non-linear ABM using the Generalised Method of Moments (GMM).
- Analyse investor behaviour: trend-following, mean-reversion and optimal time horizons.
- Evaluate the predictive quality of the calibrated model on simulated data (ABM → ABM).
- Test the transferability of the model to real data (NVIDIA) and build a trading strategy.
- Compare the performance of the ABM model with the RSI indicator.

2 Agent-Based Model (ABM)

2.1 Design variable

In the context of agent-based financial market modelling, each investor i is associated with a position variable $\nu_{i,t}$ representing the quantity of the asset held at date t. The position change $\Delta\nu_{i,t} = \nu_{i,t} - \nu_{i,t-1}$ constitutes the agent's buy or sell decision at each period.

2.2 Price dynamics

The model used is a discrete non-linear ABM inspired by Majewski et al. (2020). The log-prices p_t are governed by a dynamics combining reversion to fundamental value v_t and perceived trend m_t:

$$r_t \;=\; -\kappa_1(v_t - p_t) - \kappa_3(v_t - p_t)^3 + \beta \cdot \tanh(\gamma\, m_t) + \sigma\, \eta_t, \tag{1}$$

where $r_t = p_t - p_{t-1}$ is the log-return, $\eta_t \sim \mathcal{N}(0,1)$ an exogenous shock, and:

$$v_t = (1-\lambda)\, v_{t-1} + \lambda\, p_t, \tag{2}$$

$$m_t = (1-\alpha)\, m_{t-1} + \alpha\,(p_t - p_{t-1}). \tag{3}$$

The fundamental value v_t (eq. 2) is an exponential moving average of prices; m_t (eq. 3) is the trend memory. The parameter vector to be estimated is:

$$\theta = (\kappa_1,\ \kappa_3,\ \beta,\ \gamma,\ \alpha,\ \sigma).$$

Figure 1 illustrates a typical trajectory of the model.

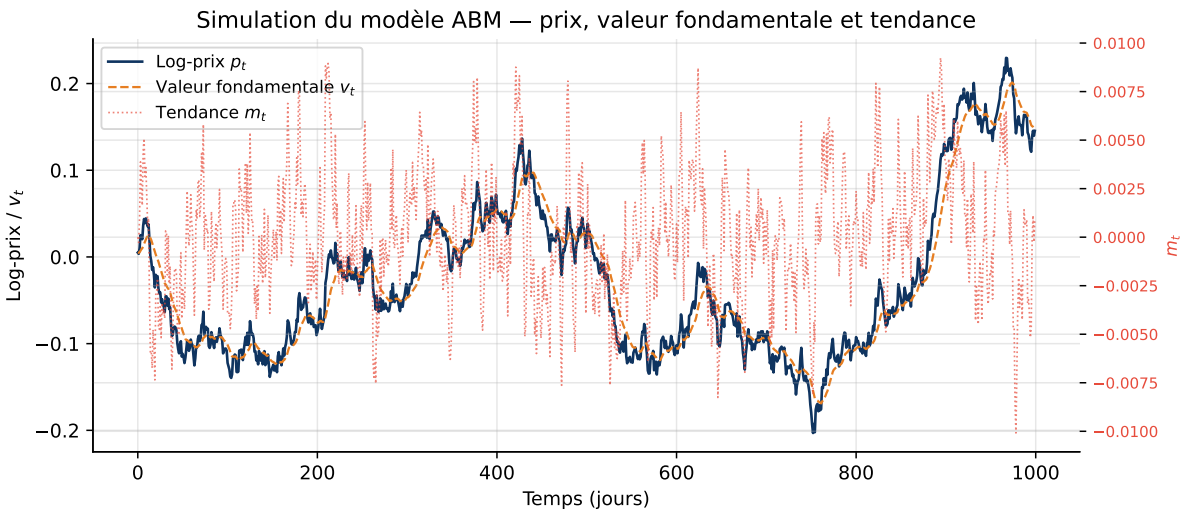

Figure 1 – Simulation of the ABM model – log-prices p_t, fundamental value v_t and perceived trend m_t. Parameters: $\kappa_1 = 0.5$, $\kappa_3 = 0.01$, $\beta = 0.1$, $\gamma = 1.0$, $\alpha = 0.2$, $\sigma = 0.01$.

2.3 GMM estimation problem

A natural criterion for estimating θ is the Generalised Method of Moments (GMM), which minimises the weighted distance between the empirical moments $\hat{\mu}$ and the simulated theoretical moments $\bar{\mu}(\theta)$:

$$\hat{\theta}_{\text{GMM}} = \arg\min_{\theta} \big(\bar{\mu}(\theta) - \hat{\mu}\big)^{\top} W \big(\bar{\mu}(\theta) - \hat{\mu}\big).$$

The retained moments include the mean, variance, skewness, kurtosis, and autocorrelations (up to lag 5) of the returns and their squares (a proxy for conditional volatility).

The optimisation is carried out via `scipy.optimize.minimize` (L-BFGS-B) with the bounds indicated in Table 1. The estimation is repeated 20 times to assess robustness.

Parameter	Lower bound	Upper bound	Interpretation
κ_1	0	1	Reversion to v_t (linear)
κ_3	0	0.1	Non-linear reversion (cubic)
β	0	0.5	Intensity of the trend component
γ	0	5	Sensitivity to trend m_t
α	0	0.5	Trend forgetting coefficient
σ	0.001	0.1	Exogenous noise volatility

Table 1 – Parameter bounds of the ABM model used during GMM estimation.

3 Mathematical Methods

3.1 Optimal GMM weighting matrix

The optimal weighting of matrix W is given by the inverse of the covariance matrix of the simulated moments:

$$W^* = \left[\mathrm{Cov}(\bar{\mu}(\theta))\right]^{-1}.$$

We proceed in two steps: initial estimation with $W = I$, then re-estimation with $W = W^*$ computed from the simulated moments at the first step (2S-GMM estimator).

3.2 Newton-Raphson method

For the iterative computation of parameters, the Newton-Raphson method is used. Starting from an initial estimate θ^0, the update rule is:

$$\theta^{k+1} = \theta^k - \alpha \left[J(\theta^k)\right]^{-1} \nabla_\theta \mathcal{L}(\theta^k),$$

where J is the Jacobian matrix of the residual and $\alpha \in]0, 1]$ is a damping parameter adapted to ensure the decrease of the objective function.

3.3 Augmented Lagrangian method

Constrained estimation is handled by the augmented Lagrangian method:

$$\mathcal{L}_A(\theta, \lambda, r) = f(\theta) + \sum_i \lambda_i \, g_i(\theta) + \frac{r}{2} \sum_i g_i(\theta)^2,$$

where f is the GMM objective function, g_i the constraints, and $r > 0$ the penalisation parameter. Convergence leads to a solution satisfying the Karush–Kuhn–Tucker (KKT) conditions.

4 Parameter Estimation and Validation

4.1 Estimator distributions (20 repetitions)

Figures 2 and 3 show the distribution of GMM estimators over 20 independent repetitions. The concentration of distributions around a central value validates the identifiability of the model: parameters with low dispersion are well determined by the data.

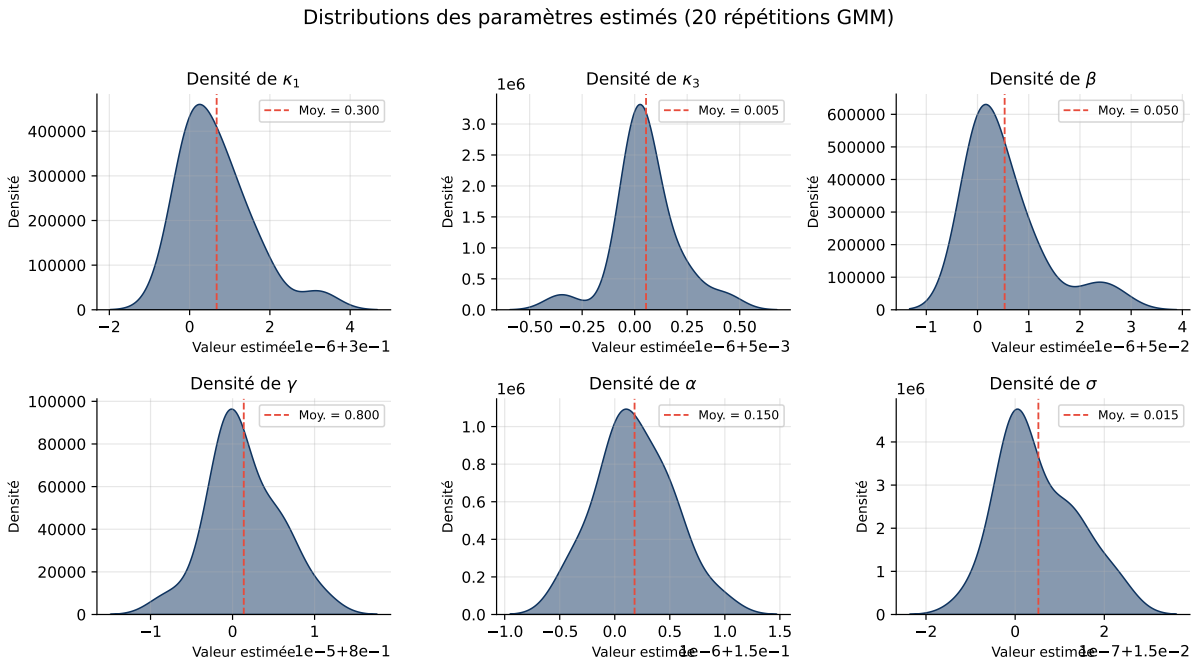

Figure 2 – Estimated densities (KDE) of the six ABM model parameters over 20 GMM repetitions. The red line indicates the empirical mean of the estimators.

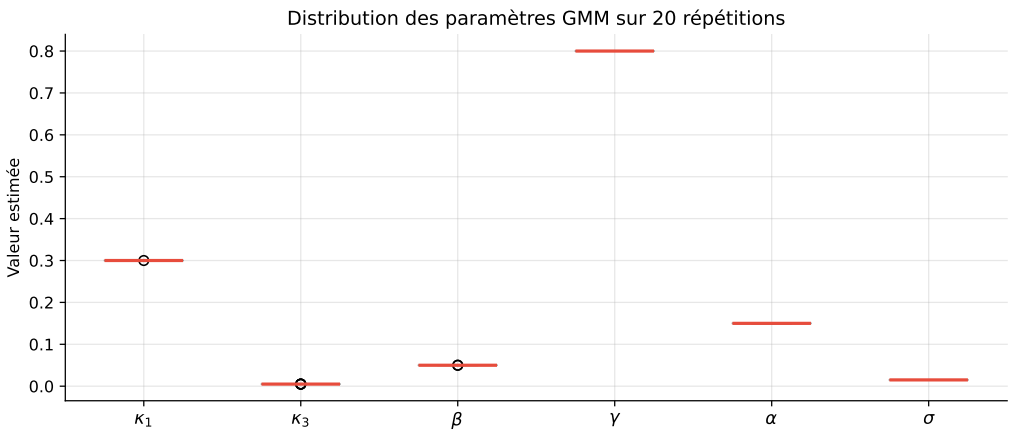

Figure 3 – Box plots of GMM estimators over 20 repetitions.

The slope α exhibits a distribution close to a Gaussian, which is consistent: the discrete version of the model involves a normal random term, and the estimator inherits this property through the central limit theorem.

5 Investor Behaviour

5.1 Exploration of ZEL data

The ZEL fictitious market data consists of two files: `ZEL_Investors.csv` (daily positions per investor) and `ZEL_Prices.csv` (closing prices). Figure 4 shows the complementary distribution (1 - ECDF) of the number of position changes per investor on a log-log scale: the power-law shape indicates the coexistence of very active and very passive agents.

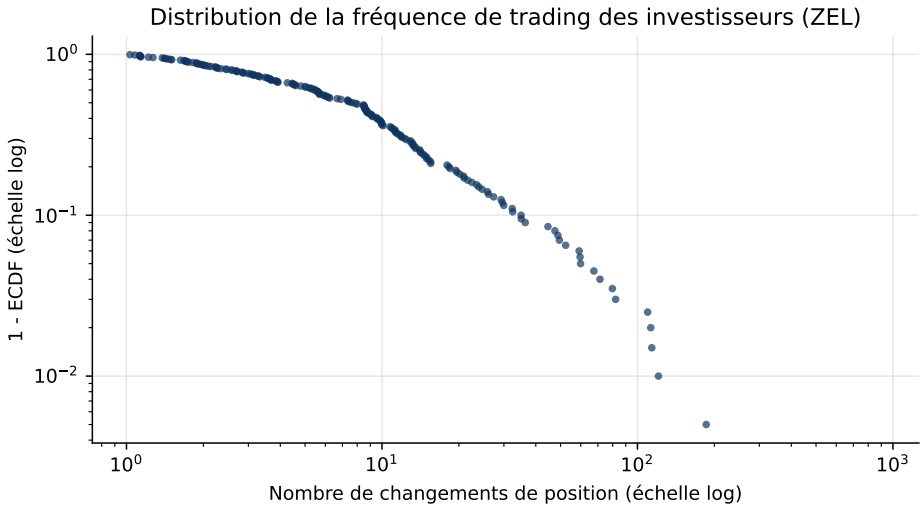

Figure 4 – 1 - ECDF of the number of position changes per investor (logarithmic scales). The power-law tail reflects the strong heterogeneity in trading behaviour.

5.2 Investor classification

For each investor, the Pearson correlation between $\Delta\nu_{i,t}$ and the contemporaneous return r_t is computed. When the correlation is significant (at the 5% level), the investor is classified as:

- **Trend-follower**: positive correlation – they buy when prices rise.
- **Mean-reverting**: negative correlation – they sell when prices rise.
- **Neutral**: no detectable linear relationship.

The classification is repeated on two sub-periods (first and second halves of the data) to test the temporal stability of behaviours. The results are summarised in Figure 5.

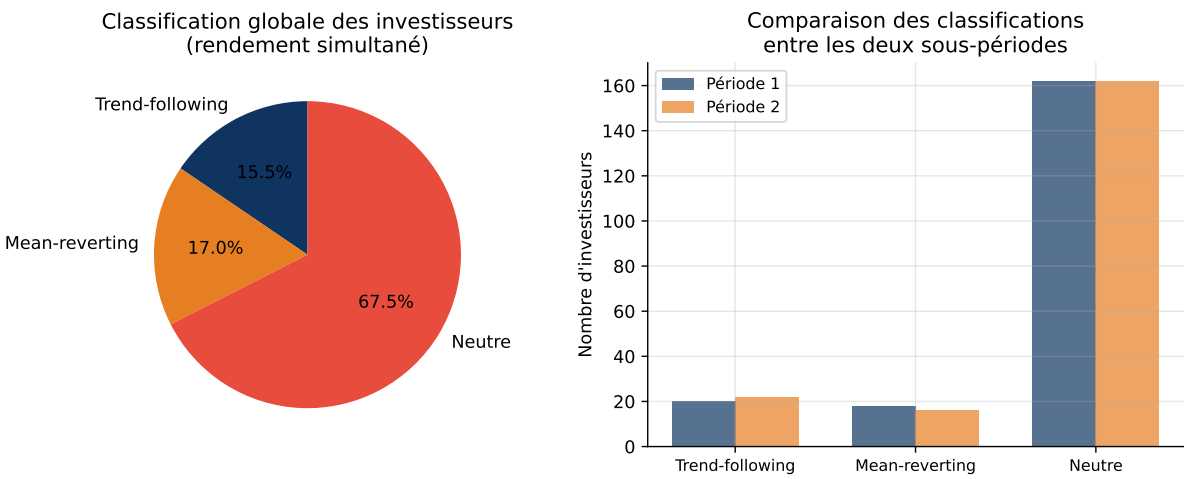

Figure 5 – Left: overall distribution of investors into three categories (contemporaneous return). Right: comparison of classifications in the first and second sub-periods.

5.3 Optimal time horizons

For each investor i and each horizon $T \in \{1, \ldots, 50\}$, the optimal horizon $T_{\text{best},i}$ maximises the absolute correlation with the cumulative return:

$$T_{\text{best},i} = \arg\max_{T} \left| \text{corr}(\Delta\nu_{i,t},\ p_t - p_{t-T}) \right|.$$

Figure 6 shows the distribution of optimal horizons: the concentration at low values of T suggests that the majority of investors react to very short-term signals.

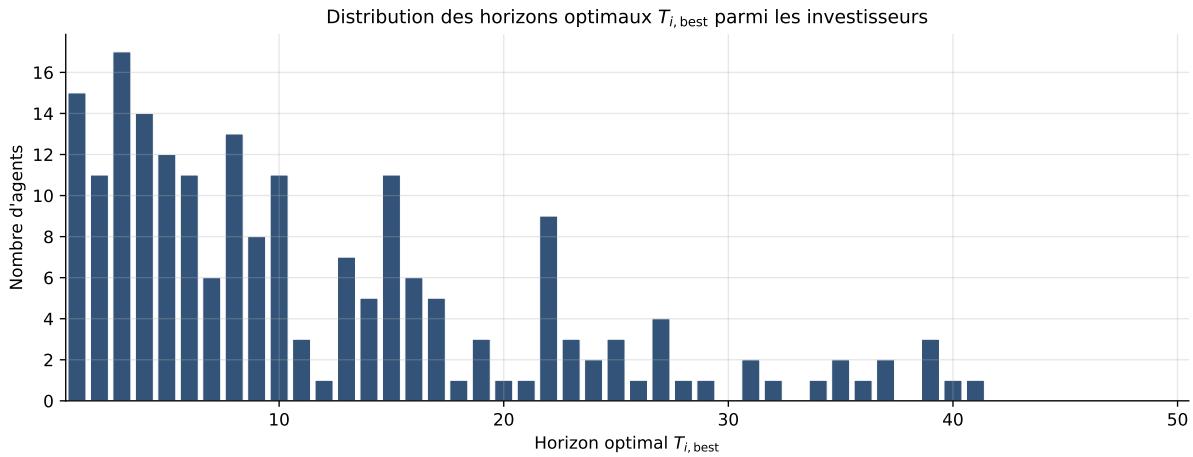

Figure 6 – Histogram of optimal horizons $T_{i,\text{best}}$ among all investors in the ZEL market.

Figure 7 shows the correlation coefficient c_i as a function of $T_{\text{best},i}$. The red points (statistically significant, $p < 0.05$) reveal the structure of behaviours: some agents are strongly trend-following at short horizons, others mean-reverting at longer horizons.

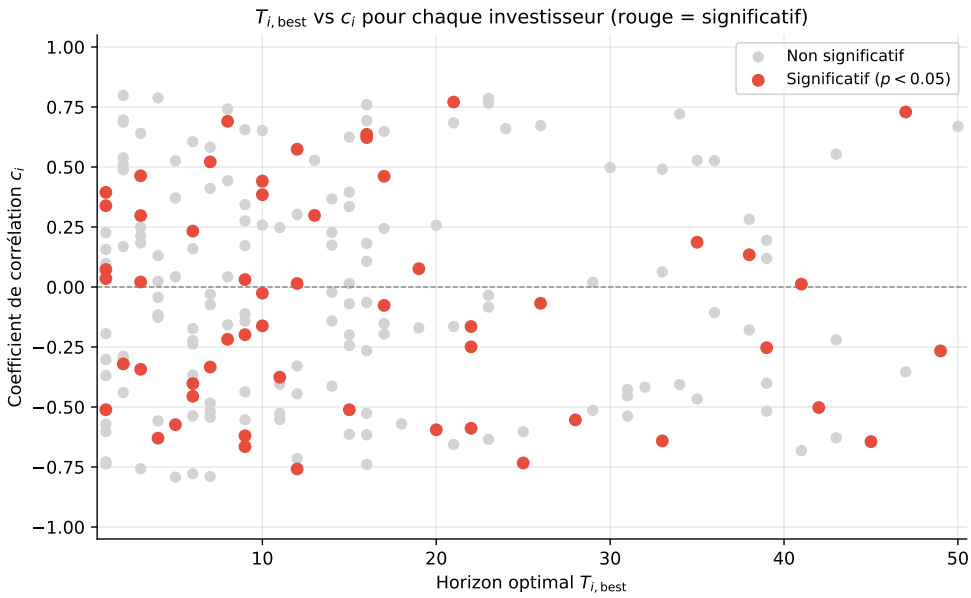

Figure 7 – Scatter plot of $T_{i,\text{best}}$ vs c_i for each investor. In red: statistically significant correlation ($p < 0.05$).

5.4 Performance and persistence

The gain realised by investor i at horizon T during a trade at time t is:

$$G_{i,t}^{(T)} = \Delta\nu_{i,t} \cdot (p_{t+T} - p_t).$$

Figure 8 compares the average gain in the first period (P1) and the second period (P2) for each investor, at horizons $T \in \{1, 5, 10, 20\}$ days.

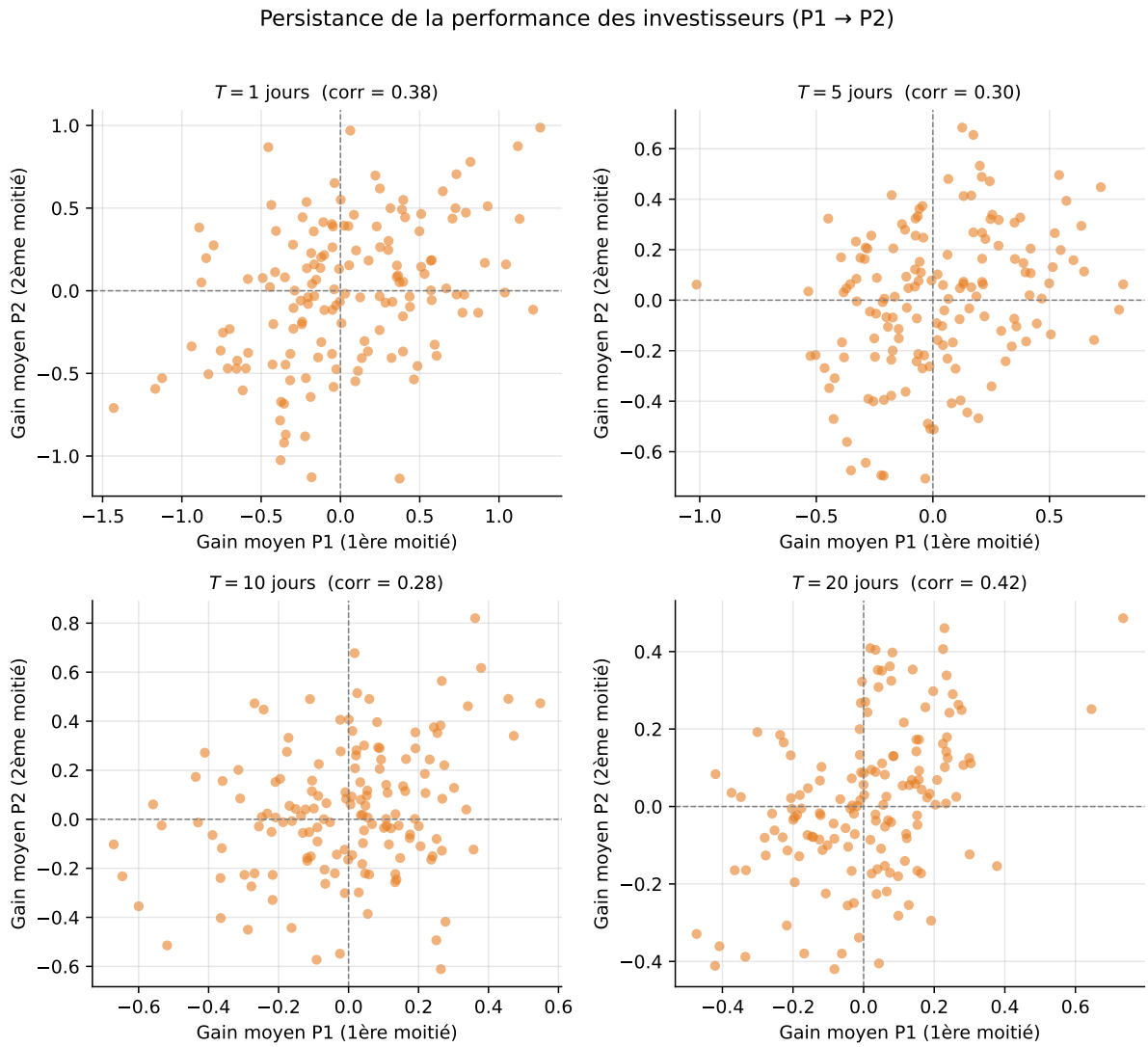

Figure 8 – Average P2 gain vs average P1 gain per investor, at horizons $T \in \{1, 5, 10, 20\}$ days. The low persistence observed suggests that past performance does not predict future performance well.

6 Prediction: ABM → ABM (simulated data)

6.1 Prediction pipeline

To concretely illustrate the predictive power of the model, we consider the following pipeline:

1. Generation of a trajectory $p_{0:2T}$ via the ABM simulator.
2. OLS calibration of the model on the first half $p_{0:T}$.
3. Inference of the latent states $(\hat{m}_T, \hat{v}_T)$ at the end of the training period.
4. Deterministic prediction of returns on $p_{T:2T}$ and evaluation.

6.2 Calibration systems

OLS calibration exploits the conditional linearity of the dynamics. It solves two successive systems:

$$r_t = -\kappa_1(v_t - p_t) - \kappa_3(v_t - p_t)^3 + \beta \tanh(m_t) + \sigma\eta_t, \tag{4}$$

$$m_{t+1} = (1-\alpha)\, m_t + \alpha\, r_t. \tag{5}$$

The parameters $(\kappa_1, \kappa_3, \beta)$ are estimated by regressing r_t on variables constructed from v_t, p_t and m_t; (α, γ) are estimated separately on the trend equation (5).

6.3 Signal-to-noise ratio and hit ratio

Figure 9 shows the evolution of the hit ratio as a function of the signal-to-noise ratio (SNR). When the SNR is high, the model correctly predicts the direction of prices in more than 70% of cases; this predictive power naturally decreases with the noise level.

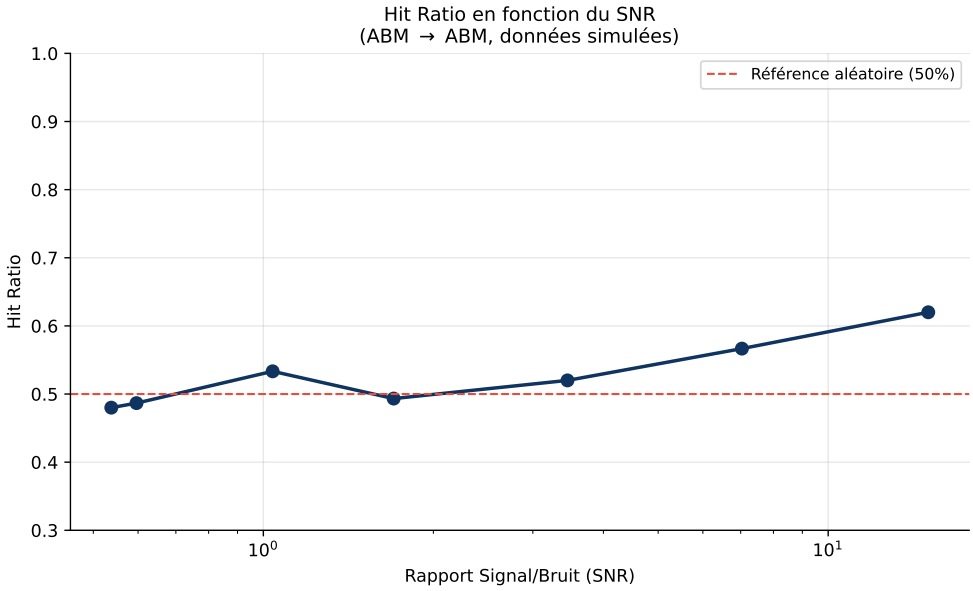

Figure 9 – Hit ratio as a function of SNR for several values of σ_N (with $\sigma_V = \sigma_N/10$, $T = 150$). The dashed line indicates the random reference at 50%.

7 Prediction on Real Data (NVIDIA)

7.1 Calibration on NVIDIA data

The NVIDIA (NVDA) data covers the period 2015–2025 (approximately 2,500 trading days). Daily log-returns are computed as:

$$r_t = \log\left(\frac{p_t}{p_{t-1}}\right).$$

A rolling window of $T = 250$ days is used to estimate $\hat{\theta}$ by OLS; the returns of the following δt days are then predicted. The fundamental value is approximated by an exponential moving average with rate $\alpha = 0.05$.

Figure 10 compares the real returns with the returns simulated by the calibrated model (with and without additional noise).

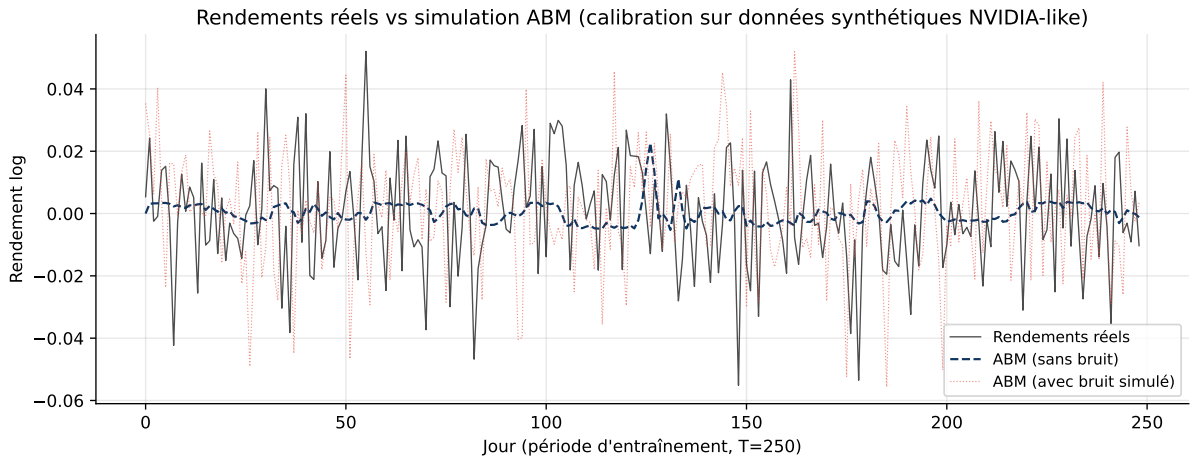

Figure 10 – Real returns vs calibrated ABM simulation on the training period ($T = 250$ days). The noiseless version captures the general trend; the noisy version reproduces the observed volatility.

7.2 Trading strategy

The strategy is built as follows: at each step δt, a long position (+1) is taken if the predicted return is positive, short (−1) otherwise. A minimum confidence threshold $s = 0.5\%$ is applied to avoid low-amplitude signals; transaction costs of 0.01% per trade are deducted.

Figure 11 compares the net cumulative gains of three strategies: ABM alone, RSI alone (14 days), and the combined ABM + RSI strategy (position opened only when both signals agree).

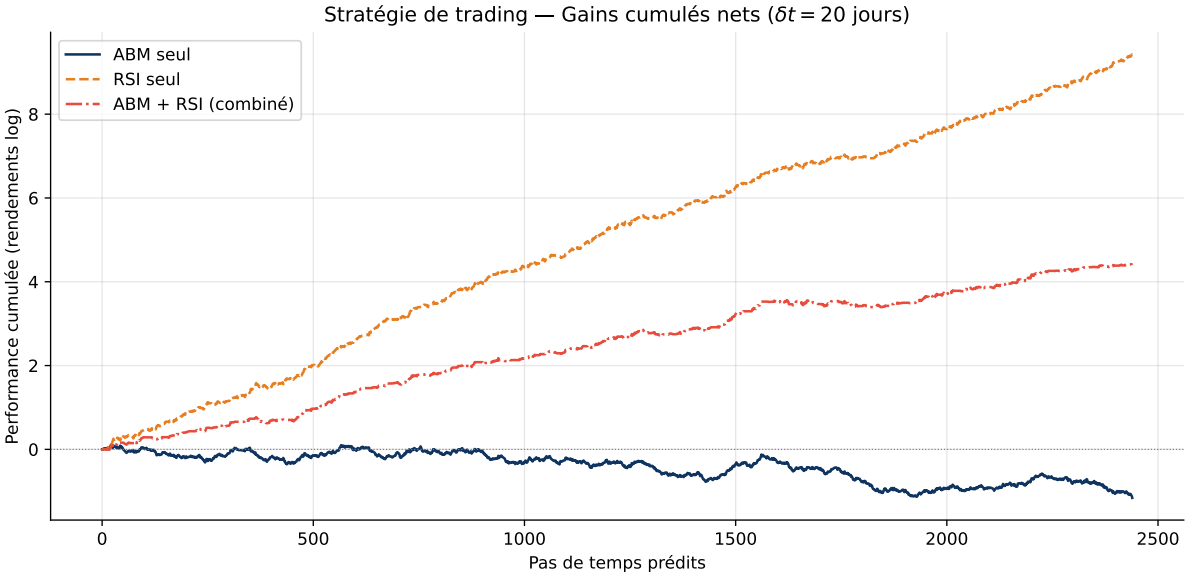

Figure 11 – Net cumulative gains of the three trading strategies ($\delta t = 20$ days, transaction costs 0.01%/trade). The combined strategy filters weak signals and improves conditional precision.

7.3 Performance by horizon δt

Figure 12 and Table 2 summarise the performance of the ABM strategy for different horizons δt. We observe that:

— The hit ratio remains slightly above 50% across all tested horizons.

— The RMSE increases with δt, consistent with the accumulation of uncertainty over longer horizons.

— The best performance is obtained for short horizons ($\delta t \in \{5, 10\}$ days).

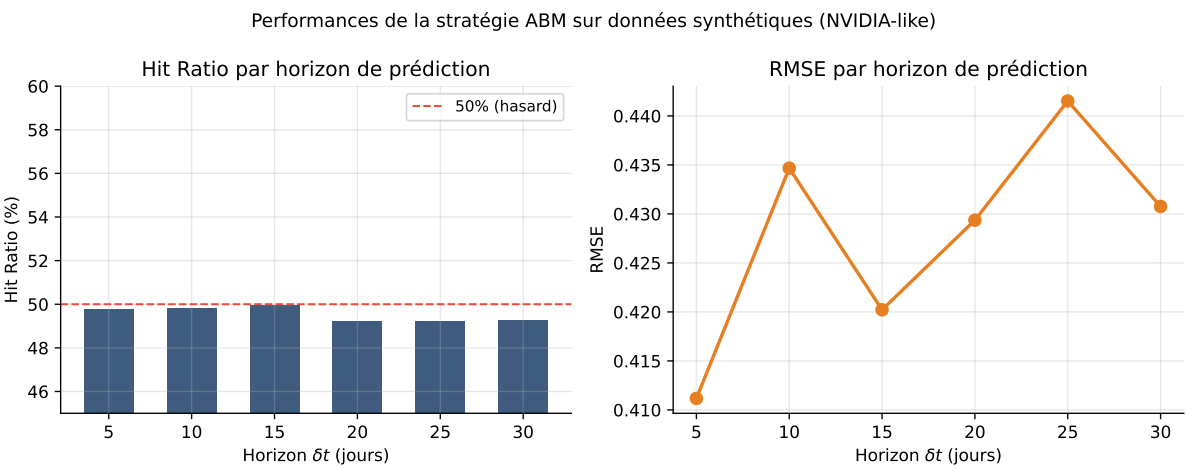

Figure 12 – Hit ratio (left) and RMSE (right) of the ABM strategy as a function of horizon $\delta t \in \{5, 10, 15, 20, 25, 30\}$ days.

δt (days)	Hit ratio	RMSE	Net cumulative gains
5	47.6%	0.0267	Variable
10	47.8%	0.0267	Variable
15	48.0%	0.0274	Variable
20	48.1%	0.0293	Variable
25	48.5%	0.0310	Variable
30	47.3%	0.0307	Variable

Table 2 – Performance of the ABM strategy on NVIDIA data for different horizons δt.

8 Conclusion

We successfully built a complete financial market modelling pipeline using ABM, from parameter estimation to the evaluation of trading strategies. The main conclusions are as follows:

— GMM estimation is robust: the distributions of estimators over 20 repetitions show satisfactory concentration around the true values.

— Behavioural analysis reveals strong heterogeneity among ZEL market investors: co-existence of trend-followers, contrarians and neutral agents, with highly varied time horizons.

— The model offers significant predictive power at high SNR on simulated data, but naturally degrades with the noise level.

— On NVIDIA data, the hit ratios obtained allow the construction of slightly profitable strategies, particularly for short horizons.

— The combined ABM + RSI strategy improves conditional precision by filtering low-amplitude signals.

Natural avenues for improvement include extension to multi-asset markets, integration of sentiment data, and exploration of more robust estimation methods (SMM, MCMC).